

MAN-CHEN LIAO

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🌐 Website

🏠 U.S. Permanent Resident

EDUCATION

National Yang Ming Chiao Tung University (NYCU)

B.S. in Electronics and Electrical Engineering

CGPA: 4.25/4.3; Last 60 GPA: 4.26/4.3; Department Ranking: 8/260

Hsinchu, Taiwan

Sept. 2024 – Present

National Yang Ming Chiao Tung University (NYCU)

B.S. in Electrophysics (transferred to Electronics and Electrical Engineering in Fall 2024)

Hsinchu, Taiwan

Sept. 2023 – June 2024

RESEARCH INTERESTS

Analog and Mixed-Signal IC Design, Data Converters, Sensor Interface Circuits

RESEARCH EXPERIENCE

National Science and Technology Council (NSTC)

Undergraduate Researcher advised by Prof. Yu-Te Liao

Hsinchu, Taiwan

July 2026 – Feb. 2027

- Awarded the NSTC College Student Research Scholarship for the project titled “An Integrated Electrochemical and Optical Smart Sensing Platform for Farm Animal Detection,” Grant No. 115-2813-C-A49-182-E (NTD 6,000/month).
- Evaluated and modified a potentiostat front-end for electrochemical sensing and conducted chip-level measurements to assess circuit and sensor responses.

Wireless Integrated Microsystems Lab, NYCU

Undergraduate Researcher advised by Prof. Yu-Te Liao

Hsinchu, Taiwan

July 2025 – Present

- Designed and optimized a two-stage operational amplifier featuring a zero-nulling resistor (R_z) for frequency compensation. Achieved a 70.17 dB DC gain, 955.6 kHz unity-gain bandwidth, and 82.96° phase margin.
- Designed and simulated an ultra-low-noise transimpedance amplifier (TIA) topology with a DC current removal block to prevent integrator saturation. Achieved a 171 dB Ω transimpedance gain and 1 MHz bandwidth, with DC-current removal verified for 0.3–10 nA input currents and power consumption reduced to 9.87 mW.

Chien Lab, UC Berkeley

Summer Research Intern advised by Prof. Jun-Chau Chien

Berkeley, California

July 2026 – Aug. 2026

- Implemented and verified a behavioral Verilog-A model of a continuous-time zoom ADC in Cadence Virtuoso, including a differential integrator, 7-bit CDAC, DM/CM smart-tracking logic, digital feedback, and $\Delta\Sigma$ modulation. Developed control timing and integrated behavioral blocks to reproduce the complete ADC conversion flow, with system-level verification through transient and spectral simulations under different input conditions.
- Studied advanced analog integrated circuits and data converters (ADC/DAC), and participated in weekly journal club meetings to discuss cutting-edge IC research.

SELECTED PROJECTS

CMOS Low-Dropout Regulator with Integrated Flash ADC

Semester Project, Analog Integrated Circuits Lab

Hsinchu, Taiwan

Mar. 2026 – June 2026

- Constructed a transistor-level power management and data conversion circuit comprising an LDO, a Bandgap Reference (BGR), and a Flash ADC. Built the error amplifier (two-stage OP) for output voltage regulation and the BGR for stable 1.2 V reference voltage, supporting digital output logic and a 100 mA load current.
- Conducted rigorous pre- and post-layout simulations to verify loop stability, transient performance, and line/load regulation. Completed DRC, LVS, and PEX using TSMC 180 nm PDK.

PUBLICATION

Man-Chen Liao and Hung-Chang Liao, “Design and Experimental Study of a Vocal Effects Unit Based on Frequency Response and Amplifier Design: A Comparison Between Practical Implementation and LTspice Simulation,” *24th Annual Hawaii International Conference on Education (HICE)*, Honolulu, Hawaii, USA, Jan. 2026.

TEACHING EXPERIENCE

Teaching Assistant, Electronics Laboratory (II)

Hsinchu, Taiwan

Lecturer: Li-Chou Chen

Feb. 2026 – June 2026

- Supported the instruction of the undergraduate laboratory course for over 50 students, helping refine course materials and prepare supporting instructional content.
- Provided hands-on experimental guidance, debugged circuit setups, and delivered concept presentations during lab sessions.

HONORS & AWARDS

Shun-I Chu Zyxel Scholarship (NTD 120,000)

Hsinchu, Taiwan

National Yang Ming Chiao Tung University

2026

Awarded to 4 junior students across the College of Electrical and Computer Engineering.

Dean's List Award (4 Semesters)

Hsinchu, Taiwan

National Yang Ming Chiao Tung University

2023 – 2025

Recognized for academic excellence, awarded to students ranked in the Top 5% of the class each semester.

The Datatronics Technology, Inc. Scholarship

Hsinchu, Taiwan

National Yang Ming Chiao Tung University

2025

Selected as the sole recipient among junior applicants in the Department of Electronics and Electrical Engineering.

NYCU Jin Qiu Bei Table Tennis Tournament

Hsinchu, Taiwan

National Yang Ming Chiao Tung University

2025, 2026

Achieved 2nd Place in Women's Singles in 2025 and 3rd Place in 2026 out of all participating competitors in the campus-wide tournament.

EXTRACURRICULAR ACTIVITIES

Hot Music Club, High School

Taichung, Taiwan

President & Keyboardist

Sept. 2021 – June 2022

- Led club operations, coordinated rehearsals and performances, and managed event planning.

NYCU EE Department Table Tennis Team

Hsinchu, Taiwan

Member

Sept. 2024 – Present

- Contributed to the team in winning the Annual Xiaomeichu Tournament, jointly organized by the EE departments of NYCU and NTHU, for two consecutive years.

NYCU Guitar Club

Hsinchu, Taiwan

Member

Sept. 2023 – Present

- Performed on stage as a vocalist, guitarist, and keyboardist in club performances.

2026 Virtual U.S.-Taiwan-Japan Semiconductor Collaboration Program

Online

Participant

2026

- Completed the program hosted by the American Institute in Taiwan (AIT), focusing on semiconductor policies, technological innovation, and talent development.

TECHNICAL SKILLS

Virtuoso, Spectre, Calibre, Hspice, Verilog-A, LTspice, Oscilloscope, Function Generator, Python, C++